

RAID (Redundant Array of Independent Disks)

RAID (Redundant Array of Independent Disks) is a data storage technology that combines multiple physical hard drives into a single logical unit for

It's used to improve **improved performance, fault tolerance, or both**.

The various ways data is distributed across the drives are called **RAID levels**, each with a different balance of reliability, performance, and capacity.

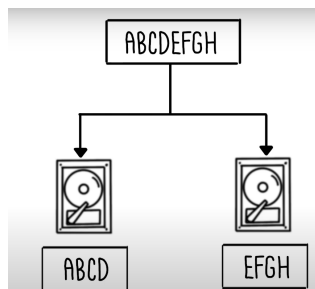
Main Purposes of RAID

1. **Redundancy (Data Protection)** – Protect data from drive failure.
2. **Performance (Speed)** – Improve read/write speeds.
3. **Capacity (Storage size)** – Combine multiple disks into one volume.

Key Concepts

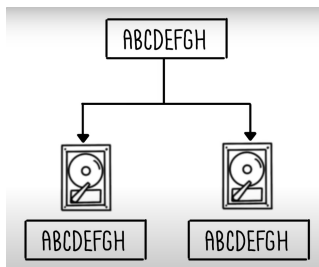
- **Striping**: Splits data across disks for speed.
- **Mirroring**: Duplicates data for redundancy.
- **Parity**: **Error-checking data** to rebuild lost data after disk failure.
- **Hot Spare**: Extra disk ready to replace a failed one automatically.
- **Controller**: Hardware/software managing RAID operations.

1. RAID 0 (Striping)



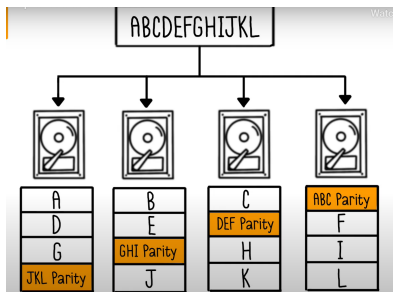
- **How it works**: Data split across multiple disks for faster read/write.
This level writes an identical copy of data to two or more drives. It provides **high fault tolerance** because if one drive fails, the data is still accessible on the mirrored drive.
- **Pros**: High performance (speed).
- **Cons**: No redundancy; one disk failure loses all data.
- **Use case**: Non-critical data needing speed (e.g., video editing).

2. RAID 1 (Mirroring)



- **How it works:** Data duplicated on two or more disks.
- **Pros:** Full redundancy; data safe if one disk fails.
- **Cons:** Lower storage efficiency (half capacity used).
- **Use case:** Critical data requiring high reliability (e.g., servers).

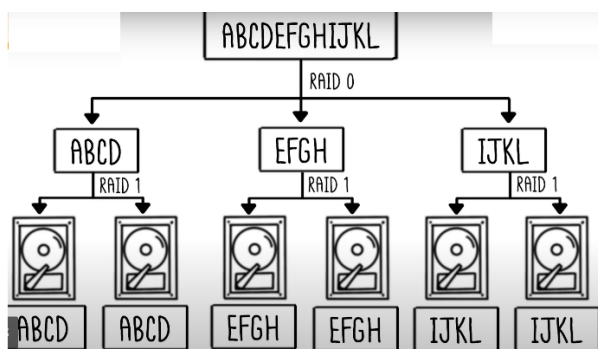
3. RAID 5 (Striping with Parity)



How it works: Data and parity info spread across 3+ disks.

- It stripes data across multiple drives and includes a parity block on a different drive. This parity information can be used to reconstruct data if a single drive fails. It requires at least three drives and can only tolerate **one drive failure**.
- **Pros:** Balances speed and redundancy; can recover from single disk failure.
- **Cons:** Slower writes due to parity calculation; rebuild after failure can be slow.
- **Use case:** General-purpose servers, cost-effective redundancy.

4. RAID 10 (1+0, Mirroring + Striping)



- **How it works:** Combines RAID 1 (mirroring) and RAID 0 (striping).

- It creates a striped array of mirrored drives. RAID 10 provides both the **high performance** of striping and the **excellent redundancy** of mirroring. It requires a minimum of four drives and can often sustain multiple drive failures, as long as they are not in the same mirrored pair.
- **Pros:** High performance and redundancy; survives multiple failures if in different mirror sets.
- **Cons:** Expensive; uses half capacity for mirroring.
- **Use case:** High-performance, mission-critical systems (e.g., databases).

Pros of RAID

- Improved performance (RAID 0, 10).
- Data redundancy (RAID 1, 5, 6, 10).
- Scalability for storage needs.

Cons of RAID

- Increased cost (more disks, controllers).
- Complexity in setup/management.
- Not a backup solution; doesn't protect against file corruption or accidental deletion.

Notes

- **RAID is not a backup solution** — it protects against hardware failure but not accidental deletion or malware.
- Always combine RAID with **regular data backups**.
- **Hardware vs. Software RAID:** Hardware uses dedicated controllers; software uses OS/CPU.
- **Backup Reminder:** RAID isn't a substitute for backups; use separate backups for data protection.
- **Choosing RAID:** Depends on needs (speed, redundancy, budget, scale).

Types of RAID Levels

RAID Level	Min. Drives	Redundancy	Performance	Description
RAID 0	2	✗ No	✓ High	Data is striped across disks. Fast but no fault tolerance.
RAID 1	2	✓ Yes	✓ Read only	Disk mirroring. Same data on both drives . High redundancy.

RAID Level	Min. Drives	Redundancy	Performance	Description
RAID 5	3	✓ Yes	✓ Good	Data + parity distributed . Balanced speed & protection.
RAID 10 (1+0)	4	✓ Yes	✓ Very High	Combines RAID 1 and 0: mirroring + striping . Fast & redundant.

When to Use Which RAID?

Use Case	Recommended RAID
Maximum speed , no data protection	RAID 0
Critical data, redundancy is a must	RAID 1
Balanced performance and fault tolerance	RAID 5
Need for extra fault tolerance	RAID 6
High performance and high fault tolerance	RAID 10

Example Use-Cases

- **RAID 0** – Gaming PCs (where performance matters, not data safety)
- **RAID 1** – Financial systems, servers (where data backup is critical)
- **RAID 5/6** – File servers, backup servers (balanced need for speed and safety)
- **RAID 10** – Databases, high-traffic applications (need speed and redundancy)

RAID 0

Best for maximum performance, with no need for fault tolerance

RAID 1

Ideal for a strong focus on high availability and data redundancy

RAID 5

Suitable for high read/write performance with a balance of fault tolerance

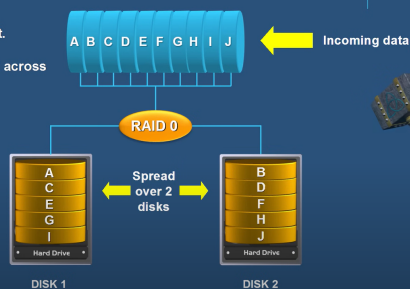
RAID 10

Perfect for high performance, low latency, and high availability

RAID 0 STRIPING

Not fault tolerant.

Data is 'striped' across multiple disks.

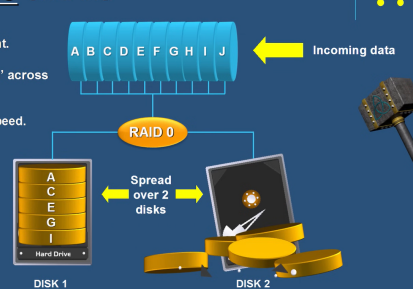


RAID 0 STRIPING

Not fault tolerant.

Data is 'striped' across multiple disks.

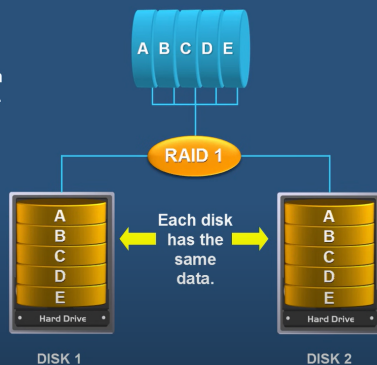
Advantage is speed.



RAID 1 MIRRORING & DUPLEXING

Is fault tolerant.

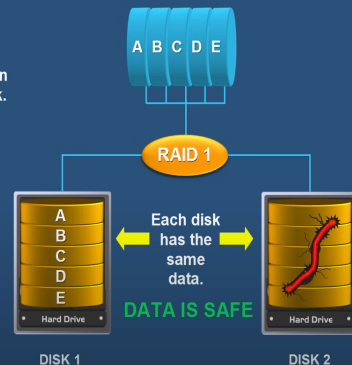
Data is copied on more than 1 disk.



RAID 1 MIRRORING & DUPLEXING

Is fault tolerant.

Data is copied on more than 1 disk.

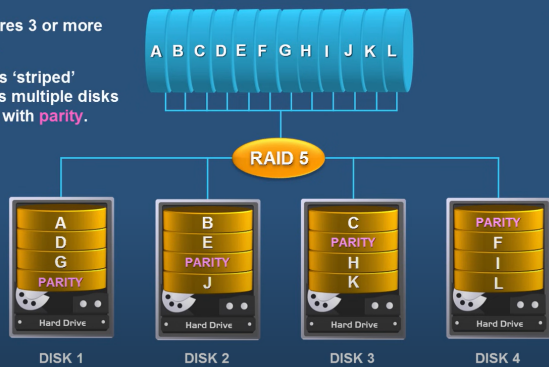


TIP: High powered lasers are not friendly to hard drives.

RAID 5 *STRIPING WITH PARITY*

Requires 3 or more disks.

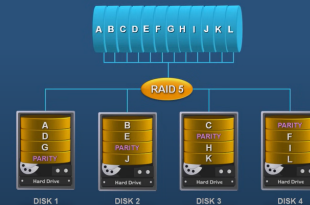
Data is 'striped' across multiple disks along with **parity**.



RAID 5 *STRIPING WITH PARITY*

The equivalent of an entire disk is used to store parity.

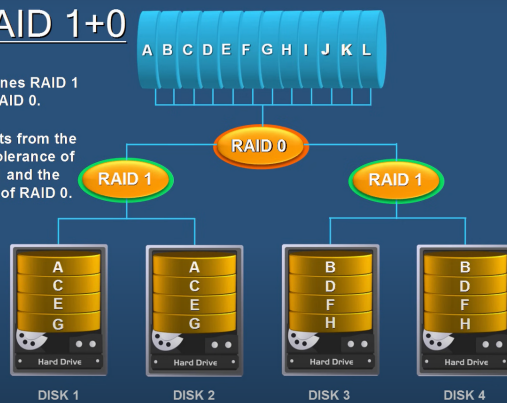
Example: An array of 4 disks totaling 4 terabytes, only 3 terabytes will be used for actual data storage.



RAID 1+0

Combines RAID 1 with RAID 0.

Benefits from the fault tolerance of RAID 1 and the speed of RAID 0.



RAID 1+0

Combines RAID 1 with RAID 0.

Benefits from the fault tolerance of RAID 1 and the speed of RAID 0.

Can only use 50% for data storage.

